Advanced Programming Practices
MTech 2026
Lab - 13

General Instructions:

- Write one .c file per problem in the assignment, renamed to RollNo_APP13_Qn.c (e.g. ABC2026001_APP13_Q1.c).
- Every program must compile without warnings using the gcc compiler.
- Free any memory you allocate. Programs that leak memory or access out-of-bounds indices will lose marks even if the visible output looks correct.
- Test each program with at least one normal case, one edge case (e.g. size 0 or 1, all-equal elements), and one larger randomly generated case.
- Submit all the .c files on Google Classroom before the due time, any late submissions will be penalized or discarded.

T is a tree if:

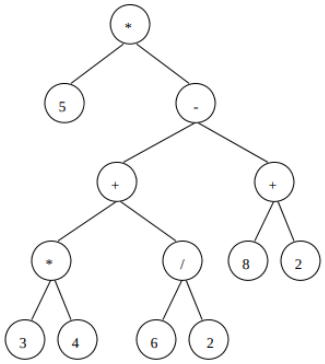
- T is empty, or
- T has a node and zero or more non-empty subtrees (each of which is also a tree.)

Binary Tree:

In a binary tree, each node has two subtrees, either of which can be empty.
we can represent such a tree with a TreeNode struct that contains two child pointers -- left and right -- like so:

```
struct TreeNode
{
   int value;
   TreeNode *left;
   TreeNode *right;
};
```

1. Using linked list representation, create a binary tree for the expression:
   $$5 * ((3 * 4) + (6 / 2) - (8 + 2))$$
2. For the tree created in Program 1, that is,



Write the program for the function(s) to traverse the tree in preorder, postorder and inorder traversal.

3. For the tree created in program 1, write a program for counting the number of leaf nodes.
4. Write a program to evaluate the expression given in Program 1, using its tree.